



		<u>Engineering Instruction 4/1</u>
<u>Title</u>		Date Issued : 01 July 20010
<u>Transformers Inspection And Test</u>		Supersedes: 29 September 2006
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1. **SCOPE**

- 1.1 The scope of the Instruction has been extended to cover the procedure to be adopted not only for export transformers up to 100kVA, but for all the ex import transformers, and also for transformers which are moved from one Division to another after having been in commission.

2. **GENERAL**

- 2.1 Since the procedure to be adopted varies to some extent with the capacity and voltage of the transformer, ex import transformers will be divided into two classes, as follows:-

CLASS A - Ex import transformers up to but excluding 500kVA capacity, with H.T. windings up to and including 11kV.

CLASS B - Ex import transformers of a capacity of 500kVA and over, including all transformers with HT windings over 11kV, irrespective of capacity.

3. **CLASS A TRANSFORMERS**

- 3.1 On receipt of any transformers of this class, Central Stores will unpack them, and then forward to Head Office a Stores Receipt Docket, which will state that the transformers concerned are ready for inspection. The buying order number under which the transformers have been supplied must be noted on the docket.
- 3.2 On receipt of this docket, a representative from Head Office will inspect the transformers externally for any sign of damage caused in transit, and in order to ensure that externally they conform to the specification.
- 3.3 One of each make and each size of transformer ordered on each Contract will be thoroughly inspected internally to ensure that the design is satisfactory, and that it conforms completely to the Specification.
- 3.4 The Head Office representative will attach a label to those transformers which are damaged, and to those referred to in paragraph 3.3 indicating that they are to be transferred to Central Workshop for inspection test and repair, if necessary.

- 3.5 Test Department will be advised of the serial numbers of the remaining transformers and they will take an oil sample from each and test for acidity only. The result of these tests will be submitted to Head Office.
- 3.6 All Class A transformers which are not damaged and which are not being internally inspected will be held in Central Stores stock and issued to Divisions and Districts as required. The ZETDC numbers and fixing screws will be firmly tied to these transformers by the Storekeepers.
- 3.7 The ZETDC numbers of those which are transferred to Central Workshops for further examination will be permanently fixed to the transformers before they are returned to Central Stores for issue.
- 3.8 A claim will be made on the Manufacturers if any damage has been sustained in transit, or if faulty workmanship is found.

4. **CLASS B TRANSFORMERS**

- 4.1 All Class B transformers will be thoroughly inspected both internally and externally, and tested shortly after they have been received, and the ZETDC numbers of these transformers will be permanently attached to them.
- 4.2 If the transformers are received fully assembled in their crates, the breather will be fitted after inspection and test and they will be inspected every three months to ensure that the breather material is still active.
- 4.3 If the transformers are received incompletely assembled, they will be carefully sealed after inspection to prevent deterioration of the oil.
- 4.4 Large transformers will invariably be shipped incompletely assembled, and, in some cases, without oil. After inspection, they will be filled with sufficient oil to cover the windings, but some space will be left in the top of the tank to allow for expansion.
- 4.5 A claim will be made on the Manufacturers if any damage has been sustained in transit, or if faulty workmanship is found.

5. **INSPECTION BY STOREKEEPER**

- 5.1 Every transformer issued to Divisions or to the Construction Department from Central Stores should be given a general external inspection by the Storekeeper, before it is despatched, to ensure that the oil level is correct and that there are no visible signs of damage.

6. **TRANSFORMERS FOR MAIN CONSTRUCTION GANGS**

- 6.1 All new transformers required by Main Construction Gangs will be fully inspected and tested at Central Workshops before being despatched, provided this service is called for on the IR's, and their ZETDC numbers will be permanently attached to them.
- 6.2 It will be the responsibility of the District Engineer in whose undertaking these transformers are installed to see that they are in good condition at the time of installation, and to carry out the inspection and tests as detailed in paragraph 9 to 13.

7. **INSPECTION AND TESTS BY DIVISION STAFF**

- 7.1 All transformers received by Divisions, irrespective of capacity, voltage, or past history, must be given the external inspection and tests as set out in Sections 9 to 13 before being commissioned.
- 7.2 All transformers received from Central Stores with their ZETDC numbers temporarily tied to them will be new transformers. If these are not fully assembled, and if assembly necessitates lifting the lid, then the cores must be raised and thorough internal inspection must be carried out in accordance with Section 8. If the transformers are fully assembled, or if assembly does not necessitate lifting the lid, then the internal inspection should not be carried out, unless the test results indicate that there is a defect in the transformer.
- 7.3 All transformers received from Central Stores with their ZETDC numbers permanently screwed to them, whether new or old, should be regarded as having been thoroughly inspected internally, and the internal inspection should not be repeated even though it may be necessary to lift the lid to assemble the transformers completely.
- 7.4 A transformer received from another Division should not be inspected internally, unless the test results are unsatisfactory, or the transformer card indicates that a period of three years or more has elapsed since it was last given a completed overhaul, in which case the internal inspection must be carried out.
- 7.5 In all cases where transformers are inspected and tested in Divisions, the test results must be forwarded to Head Office on an Inspection and Overhaul Report. Where an internal inspection is carried out, the results must be itemised on the back of the overhaul Report, in accordance with Section 8, and a remark made against each item stating that it is satisfactory or, alternatively, giving details of any defects.
- 7.6 Where an internal inspection reveals a defect in a new transformer which should be the subject of a claim on the Manufacturer, Divisions must submit their Report without delay, and mark it "URGENT - CLAIM ON MANUFACTURER".

8. **INTERNAL INSPECTION**

8.1 Having lifted the transformer core, the following points must be checked:-

- a) The core lamination iron must be inspected for signs of brittleness. If the laminations are brittle, it will probably be found that small slivers of laminae have broken away. This defect can be detected fairly easily by examining the edges of the core - the small gaps left by missing pieces will be noticeable.
- b) The core bolts and all connections must be checked for tightness.
- c) The insulation resistance between the core must be measured with a megger. If the oil is in good condition, the value obtained should be in the neighbourhood of infinity; but any value down to about 10 mega ohms will be quite satisfactory, as the only purpose of this insulation is to break what would otherwise be a circular electrical path, which would permit the flow of quite large eddy currents in the core. Any value below about 10 mega ohms will indicate either that there is considerable moisture present, which can be confirmed from the result of the oil test, or that the insulation is faulty. This test is to be carried out with a 500 volt megger, as a higher voltage might damage the insulation.
- d) The winding spacers and coil calpms or wedges must be examined to ensure that they are firmly in position.
- e) The windings must be carefully scrutinised for any signs of abrasion.
- f) The tap changing switch must be examined and tested to ensure that good contact is being made on all tap positions, and that all nuts and bolts are secure, and contacts clean.
- g) Throughout the inspection, a careful watch must be kept for any traces of small particles of metal (such as iron or copper filings, or solder) on the horizontal surfaces of the windings, the core, or the supporting structure. The bottom of the tank must also be examined for any such particles.

If, as a result of this internal inspection, it is found that the laminations are brittle, that the core bolt insulation is down, or that there is a deposit of metal particles on the horizontal surfaces, or that there is any other defects which is the result of faulty workmanship in the assembly the transformer or a weakness in design, then a detailed report should be submitted to Head Office promptly. The transformer must not be repaired and put into commission until authorisation has been received from Head Office.

9. **EXTERNAL INSPECTION**

9.1 A general external inspection shall be carried out on all transformers, particular attention being paid to the following points:-

- a) The tank and all gaskets to be checked for leaks.
- b) The bushings to be carefully examined for cracks.

- c) The oil level gauge to check for satisfactory operation, particularly if it is of the dial and pointer type operated by a float.
- d) Then oil level to be checked and the transformer topped up if necessary.
- e) The tapping switches to be tested for correct mechanical operation.
- f) The breather to be examined. If it is of the silica gel type, the breather material must be dried out or replaced if the colour indicates the present of moisture. If the transformer has been shipped in oil, and the breather is of the oil filled type, a seal will be found between the breather and the tank. This seal must be removed.

10 **MEGGER TESTS**

10.1 The following megger tests using a 2 500 volt mega will be taken, with the transformer in oil after assembly:-

- a) HT to LT
- b) HT to Earth
- c) LT to Earth

10.2 The table below is a guide to the minimum mega test results which should be obtained. The values quoted below in column 2 are mega values for HT to Earth, and to LT, and those in Column 3 are for LT to Earth.

The oil temperature at which these mega tests are taken should be recorded on the Overhaul Report.

Oil Temperature	Minimum meter Reading for 11kV Windings HT to Earth and to LT	Minimum Megger Reading for LT Windings LT to Earth
40°C	176 mega ohms	17.6 mega ohms
30°C	352 mega ohms	35.2 mega ohms
20°C	704 mega ohms	70.4 mega ohms

11. **OIL TESTS**

11.1 Oil tests are to be conducted as far as possible in accordance with BSS 148/1933 and in order to do this the following procedure is laid down.

11.2 The containing vessel and electrodes shall be made absolutely clean and dry before the oil is tested therein, and it has been found advantageous to carry out the final operation of removal of dust and fibres electrically, as stated in paragraph 11.3.

11.3 The containing vessel is filled with oil and about 20kV applied between the electrodes for 30 seconds to attract dust and fibres. The electrodes are then washed with oil and the oil in the containing vessel poured away. This operation should be carried out four times, after which the container thus cleaned may be used for twenty tests without being wiped with any fibrous material.

11.4 The tests shall be commenced at a voltage of about 10kV which shall be increased to 40kV as quickly as possible and maintained at this voltage for one minute.

- 11.5 Oil which stands 40kV for one minute is to be regarded as good oil, but it must be realised that complete breakdown is known by the establishment of an arc. Preliminary transient sparking which does not develop into an arc may be ignored.
- 11.6 Three oil samples will be taken from the transformer drain valve and flash tested.
- 11.7 The oil should be carefully drained from the transformer to avoid the formation of air bubbles, and should be allowed to stand for a few minutes before testing to give any air bubbles which have formed a chance to rise to the surface.
- 11.8 Oil shall be tested in a standard oil testing set, three tests being sufficient to classify the oil as follows:-
- a) Oil which withstands 40kV for one minute in all three tests, or in two out of three tests, is deemed to be in good condition.
 - b) Oil which does not withstand 40kV for one minute in two or three tests shall be filtered as soon as possible and further samples tested.
- 11.9 The oil should be at ambient temperature when the tests are made, and this temperature should be recorded on the overhaul Report.
- 11.10 The above value to which the oil must be flash tested applies only to new transformers and does not supersede the routine instructions given in Engineering Instruction No. 10/1.

12. **INTERPRETATION OF MEGGER AND OIL TESTS**

- 12.1 Should the megger and oil tests results be above the values detailed in Sections 10 and 11, and the transformer can be assumed to be serviceable.
- 12.2 Should the megger readings be satisfactory but the oil flash tests low, then it can be assumed that the oil contains moisture and/or foreign matter, and it must be filtered.
- 12.3 Should the megger readings be low but the oil satisfactory, it may be assumed that there is moisture trapped in the windings or in the accessories. The Transformer requires drying and, on receipt of the test results from Divisions, Head Office will issue further instructions. Divisions and Undertakings will not dry out transformers without having received instructions from Head Office.
- 12.4 When both megger and oil test results are low it will be necessary to dry out both the oil and the core, and detailed instructions will be given by Head Office, as referred to in paragraph 12.3 above.

13. **VOLTAGE TESTS**

- 13.1 In addition to the megger and oil tests detailed above, voltage tests must be taken on each tapping of the transformer and the results submitted to Head Office. This tests should be made either at the Undertaking Workshops, or after the transformer has been installed at a substation, but before a consumer is connected. Where the voltage on the LV side of the transformer is greater than 550 volts, the tests will have to be carried out with a potential transformer.

14. **ALLOCATION OF CHARGES**

- 14.1 All work has to be carried out on a transformer transferred from another Divisions shall be charged to Job 165 of the Division from which it has been transferred.
- 14.2 Work on an ex import transformer shall be charged to the Vote to which the transformer has been reserved.
- 14.3 Time and material spent on an ex import transformer which should be recoverable from the manufacturers will be charged to a special job number Divisions will be advised of the Job number to which charges are to be allocated each of them they submit an Overhaul Report marked "URGENT - CLAIM ON MANUFACTURERS".